

TECHNICAL REPORT

German Grid-Load Forecasting

Daily and hourly day-ahead forecasts

The system combines a daily forecast and an hourly profile. It uses hierarchical reconciliation and conformal prediction intervals. MLflow stores the complete reconciled model.

Summary

The system forecasts realized German electricity grid load. It produces daily and hourly day-ahead forecasts.

Release results

- Period: 01 Jan 2026 to 04 Aug 2026; n=216 days.
- Daily forecast: Ensemble. MAE 1003.6 MW, MAPE 1.867%, and MASE 0.445. Reference models: Naive (t-1) MAPE 7.025% (n=216); Seasonal naive (t-7) MAPE 5.575% (n=216).
- Hourly forecast: Residual hybrid. Daily-total alignment reduced MAE from 1627.6 MW to 1557.7 MW (-4.30%). The result contains n=5183 hourly values. The daily model was LightGBM.
- Model comparison: Validation MAE was 1465.3 MW for Residual hybrid and 1471.0 MW for Direct LightGBM. The paired difference on the final data was +1.9 MW. Its 95% range was [-48.8, 52.1] MW across 216 days. This range includes zero, so the data do not show a clear winner.
- Uncertainty ranges: Symmetric 90%: target 90%, measured coverage 84.89%, mean width 5573.1 MW, interval score 8831.6 MW, n=5183; Adaptive 90%: target 90%, measured coverage 89.95%, mean width 6529.3 MW, interval score 8250.5 MW, n=5183; CQR 90%: target 90%, measured coverage 86.51%, mean width 7550.2 MW, interval score 10567.7 MW, n=5183.
- Release check: source 555cfd0381a880ca22264e98fd94aa4004efec7e; daily protocol b6f14a50c9327083ebfb68402a7348d02a hourly protocol 2d1ac058d4b3b4a9e84365f290b52ddacd237a5624a7bfd9410ed16be0b75ff7; bundle 847a190df672fac4c6730
- Scope: The final period was already examined. These results describe this data period and do not state future accuracy.

1 Data and evaluation

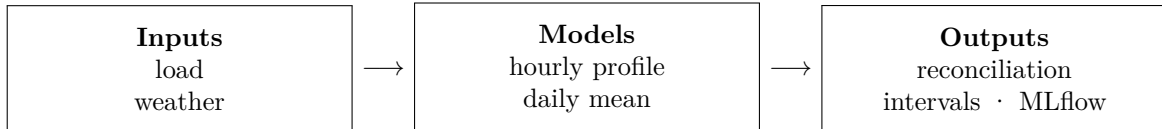
SMARD supplies realized grid-load data. Open-Meteo supplies archived weather forecasts.

The target is realized electricity consumption in MW. Residual load is outside the report scope.

Weather features use forecasts from 24 hours before valid time. Earlier data uses previous-day observed weather.

The configured chronological stages are training, validation, calibration, and retrospective final. Validation selects models before later evidence is inspected.

2 Forecast architecture



The hourly evaluation compares residual hybrid, direct LightGBM, and direct Ridge. Each model predicts valid hours directly.

The features include horizon, lag-24, lag-168, calendar, weather, and Fourier terms. Direct prediction prevents recursive error propagation.

Validation MAE selects the daily anchor. The final model uses only data before calibration.

3 Reconciliation

For local day d and hour h , the system uses:

$$\hat{y}_{d,h}^{rec} = \hat{y}_{d,h}^{hourly} \times \frac{\hat{y}_d^{daily}}{\frac{1}{|H_d|} \sum_{j \in H_d} \hat{y}_{d,j}^{hourly}}.$$

The operation keeps the relative hourly shape. The reconciled mean equals the daily forecast.

The code supports complete local days with 23, 24, or 25 hours. It rejects invalid predictions and incomplete days.

The generated release evidence reports raw and reconciled MAE from the same retrospective-final forecast rows.

4 Model selection

The release keeps the validation-selected model. Later candidate comparisons are retrospective evidence and do not change that selection.

The paired bootstrap uses one MAE difference for each local day and resamples the paired day blocks. The generated release evidence reports its result.

5 Prediction intervals

The point, daily, and quantile models use data before calibration. The models stay fixed during calibration and retrospective-final evaluation.

Reconciliation occurs before score calculation. Both interval methods use the same reconciled point forecasts.

The current implementation keeps all models frozen. Coverage is empirical; adaptive intervals are prequential monitoring without an unconditional time-series guarantee.

6 Controls

6.1 MLflow

`ReconciledForecaster` contains both fitted models and both feature schemas. It also contains the reconciliation operation.

Artifact replay is checked against the persisted evaluation predictions.

6.2 Data validation

Load	Values must be finite and positive.
Weather	Values must be within configured physical ranges.
Time	UTC timestamps must be unique, complete, and on the hourly grid.
Telemetry	Prediction and interval indexes must match before the join.

6.3 Datetime resolution

Raw `.asi8` values depend on the pandas storage unit. The feature code uses `Timedelta(hours=1)` to calculate elapsed hours.

A regression test uses `datetime64[us]`. It verifies the time scale and the weekly Fourier phase.

7 Verification and limits

```
uv run ruff check src scripts tests
uv run mypy src
uv run pytest --cov=loadfc --cov-report=term-missing --cov-fail-under=80
uv run python scripts/validate_results.py
```

Five cities represent national weather exposure. Distribution shift can change conformal coverage.

A production service requires scheduling, access control, service-level objectives, and rollback rules.

References

- Bundesnetzagentur, SMARD: <https://www.smard.de/en>
- Open-Meteo Previous Runs API: <https://open-meteo.com/en/docs/previous-runs-api>
- Hyndman et al., “Optimal Combination Forecasts for Hierarchical Time Series”.
- Romano et al., “Conformalized Quantile Regression”.
- Gibbs and Candès, “Adaptive Conformal Inference Under Distribution Shift”.
- Ke et al., “LightGBM: A Highly Efficient Gradient Boosting Decision Tree”.